Difficulties in basic concepts of mathematics in higher education: a systematic review

Priscila da Silva Neves Lima
Universidade Federal de Goiás
Goiás, Brazil
priscila_lima@ufg.br

Laira das Almas Silva
Universidade Estadual do Sudoeste da Bahia
Bahia, Brazil
lairaalmas@gmail.com

Igor Moreira Félix
Instituto de Matemática e Estatística
Universidade de São Paulo
São Paulo, Brazil
igormf@ime.usp.br

Leônidas de Oliveira Brandão
Instituto de Matemática e Estatística
Universidade de São Paulo
São Paulo, Brazil
leo@ime.usp.br

***Abstract*—This full paper presents the process and the results of a systematic review about students' difficulties in basic mathematics concepts in higher education. The main goal of this review is to identify the basic mathematics topics in which students face most difficulties. This tough relationship with mathematics begins a long time before the university, it can be identified at the first levels, during basic education, when students are introduced to arithmetic. According to studies, in most cases, there are two kinds of problems that can explain this learning difficulty: the first one is identified when it is related to the child's cognitive development, while the second kind of common learning difficulty is related to problems outside the child or in another particular problem. Also, it has been identified in the literature that when studying mathematics, a range of feelings can be triggered in students, ranging from positive feelings to negative ones. This perception can explain why students with a history of failures in mathematics believe that they are not able to construct the necessary knowledge to pursue higher education in STEM (science, technology, engineering and mathematics) courses. In Brazil, for instance, only 17% of students apply for a STEM course, while in countries in North America and Europe, this percentage reaches 24%. Therefore, this systematic review was conducted in order to analyze publications regarding difficulties related to basic math, faced by students in higher education. The results indicate various difficulty levels in a variety of mathematics topics, including functions and fractions in practical situations. The analyzed papers also point to the fact that the problems are faced when students are in the primary education and persists in higher education.**

Keywords: ***Difficulties, Fundamentals, Higher education, Mathematics, STEM education, Systematic review*.**

I. Introduction

Mathematics is an essential knowledge in modern society, allowing better problem-solving skills in different areas, from medicine to philosophy. In its concepts lay the foundation of the information system, largely used, nowadays, in machine learning, networks and mobile applications just to mention a few. Unfortunately, in many countries, the literature points to a severe deficiency in students' knowledge of basic mathematics. This problem starts at the early years of formal education, producing a high rate of people suffering from math anxiety [1].

Mathematics is directly involved in the process of developing society. The comprehension of elementary topics in mathematics builds a strong foundation to complex subjects. Automatic computational decision, for instance, is based on algorithms that perform a set of basic tasks, such as proportion and statistics. Therefore, is important to examine the role of mathematics in today's education, especially in STEM (science, technology, engineering and mathematics) education.

A deficiency in mathematical knowledge seems to be present in students at all levels of education. However, when this problem is not remedied at earlier stages of education, it can lead to great difficulties afterwards. This could even prevent people to follow a STEM professional career.

The shift in paradigm experienced in the transition from k-12 to higher education in mathematics seems to take students by surprise. They often feel unprepared, which leads to insecurity. Perhaps this is due to the fact that basic mathematics is often represented as a long succession of facts to be memorized and reproduced [2]. This could make it seem like a magical procedure, destitute of meaning, which tends to be perceived as something uninteresting and incomprehensible to many students.

This is one of the factors influencing the failure in learning mathematics. In addition, often, in higher education, computer applications are presented as a supporting tool to improve the math abstraction. These software can, for example, help students to perform long calculations, allowing them to be focused on the bigger picture and to see tools as means to problem solving [3].

The difficulties that come with studying mathematics can invoke a range of feelings on students, from positive to negative [4], becoming, sometimes, pathological and producing the previous quoted math anxiety [1]. This perception can explain

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why students with a history of failures in mathematics believe not to be able to build the knowledge needed to pursue higher education in STEM courses.

In Brazil, for instance, only 17% of students apply for STEM courses, while in countries in North America and Europe, this percentage reaches 24% [5]. Nevertheless, according to [6] engineering degrees, alongside mathematics and physics are among the courses with higher evasion rates.

Assuming the poor mathematical background to be a variable contributing to higher failure rates in STEM courses, the purpose of this work is to analyze the literature, performing a SRL (Systematic Review of the Literature) based on publications regarding math difficulties faced by higher education students.

This paper is structured as follows: section II presents the background, contextualizing the research; section III presents the protocol of the SRL, including Planning, Conducting and Reporting; Planning presents the four research questions, the selected databases and the inclusion, exclusion and quality criteria; Conducting presents the strings formatted according to each database standards and the list of selected papers to perform the SRL; Reporting presents the review in the form of answering the research question; section IV presents some considerations regarding the research; section V presents the conclusion to this review; section VI presents acknowledgements.

II. Background

This tough relationship with mathematical concepts begins at the first levels, during basic education when students are introduced to arithmetic. According to studies [7], in most cases, there are two kinds of problems that can explain this learning difficulty: the first one is intrinsic, which means it is related to the child's cognitive development, while the second is extrinsic, related to problems outside of the child.

Some researches claim that mathematical subjects that are actually used or needed in real life situations and "school mathematics" are not related [2], [8]. This distance between them could be another reason it is considered difficult or non-attractive to young learners. This paints an unpleasant picture, since the lack of adequate domain of mathematics appears to be a possible factor for the lack of interest in STEM education and, consequently, their high dropout rates, especially in computer science and engineering undergraduate courses [9].

The literature available focusing on challenges faced by higher education students regarding basic mathematical concepts [10], [11] often focus on demographic data or students scores. Most of them, however, do not point to specific concepts these students seem to be finding most difficult. In this context, a SRL is proposed, aiming to identify which are these basic mathematical concepts that students face most difficulties in higher education.

This SRL is based on the methodology proposed by Kitchenham [12], that follows three stages: planning, conducting and reporting. The planning stage consists of identifying the need for a review and defining a protocol. The conducting stage includes identifying research, selecting studies, evaluating studies, extracting data, and ultimately synthesizing data. Lastly, the reporting stage is one-step only.

III. Method/Protocol

This research was conducted in order to review publications on the topic of mathematical difficulties encountered by students attending higher education. A selection criteria was developed to identify the primary and secondary studies within the available researches on the topic, aiming to answer to the general research question:

In which basic mathematical concepts students in higher education face most difficulties?

A. Planning

In order to guide the research, the following questions were defined:

1) Research Questions:

- **Q1.** Which mathematical topics are most frequently pointed as hard to learn?
- **Q2.** Which knowledge subarea is impacted by this deficiency in fundamental mathematical topics?
- **Q3.** What are the approaches that can help to minimize these high failure rates?
- **Q4.** What are the technologies proposed to enhance the mathematical learning process?

2) Databases: This study was conducted on three consolidated databases in the scientific scope: ACM Digital Library (dl.acm.org), IEEE Xplore Digital Library (ieeexplore.ieee.org) and Scopus (www.scopus.com).

3) Inclusion Criteria: Were added to the review, primary studies within the following criteria:

- **I1.** Published from 2009 to 2019;
- **I2.** Published in English or Portuguese;
- **I3.** Publications containing "math" in at least one of the author's keywords.

4) Exclusion Criteria: Were removed from the review, primary studies within the following criteria:

- **E1.** Duplicated papers;
- **E2.** Not accessible online;
- **E3.** Not a book chapter, article or conference paper.
- **E4.** A threat to method validity;
- **E5.** Methodologies or final results.

5) Quality Criteria: Were only considered publications that approached the subject of difficulties in basic mathematical concepts in higher education in a way to be able to provide answers to the research questions.

B. Conducting

To execute this SRL, a three phase strategy was used to refine the search string. Initially the string was formed using keywords based on the general research question. This string were, then, formatted according to each database and used in its correspondent search engine, which resulted in a large

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number of records (ACM: 23, IEEE Xplore: 825, Scopus: 601). To evaluate the quality of the search string, the results were randomly selected and analyzed, however, in the end, the interest set was empty. Then, an ad-hoc search was used, returning papers. The second and third strategy followed the same principle.

This multi-phase strategy was designed as an attempt to increase the chances of finding publications without gathering too many results out of the subject, trying not to exclude articles of interest.

These search strings and their results are as follows:

Strategy 1: Since the search is broad and can involve several areas, such as geography or biology, the search string was designed to be generic. Word variations, such as "difficulty" and "difficulties", were considered.

- ACM: (difficulty difficulties challenge challenges challenging) +(math mathematics) +("higher education" undergrad undergraduate);
- IEEE Xplore: (((difficult* OR challeng*) AND math*) AND "higher education" OR undergrad*);
- Scopus: (difficult* OR challeng*) AND math* AND ("higher education" OR undergrad*).

The application of the research protocol recovered 2,623 papers. Their title and abstract were analyzed using the selection criteria. When this was not enough, the conclusion was also analyzed. In the end, 86 papers were selected for complete analysis. The Table I shows the results with the final synthesis data extraction.

Table I
SYNTHESIS DATA EXTRACTION - STRING ONE

Database	Total	Inclusions	Exclusions
ACM	233	3	230
IEEE Xplore	561	31	530
Scopus	1,829	52	1,777
Total	2,623	86	2,537

After applying the quality criteria, 11 papers were selected, as can be seen in the Table II.

Strategy 2: The second database search was executed under the same protocol as the first, only difference being the search string, which was more specific. It used key therms to force the presence of elementary mathematics, as it is the target of this study.

- ACM: (difficulty difficulties problem fail challenge) +(mathematics mathematical) +("secondary school" "primary school" "high school" "middle school" "elementary school") +(questionnaire exam grade outcome) +(algebra arithmetic fraction decimal equation counting set);
- IEEE Xplore: ((((((difficulty OR difficulties OR problem OR fail OR challenge)) AND (mathematics OR mathematical)) AND ("secondary school" OR "primary school" OR "high school" OR "middle school" OR "elementary school")) AND ("questionnaire" OR exam OR grade OR outcome)) AND (algebra OR arithmetic OR fraction OR decimal OR equation OR counting OR set));

Table II
ARTICLES LIST - STRING ONE

Year	Reference	Title
2019	[13]	Investigating the Impact of an Immersive Computer-based Math Game on the Learning Process of Undergraduate Students
2017	[14]	Problematic topics in first-year mathematics: lecturer and student views
2015	[15]	Assumed mathematics knowledge: The challenge of symbols
2015	[16]	Mathematics diagnostic testing in engineering: an international comparison between Ireland and Portugal
2014	[17]	Making Students' Thinking Explicit: Learning What They Know about Functions
2013	[18]	Developing digital technologies for undergraduate university mathematics: Challenges, issues and perspectives
2013	[19]	Maths diagnostic testing in engineering: Dublin institute of technology and Coimbra Institute of Engineering
2013	[20]	Noticing relevant problem features: Activating prior knowledge affects problem solving by guiding encoding
2010	[21]	Diagnosing the problem: Using a tool to identify pre-registration nursing students' mathematical ability
2010	[22]	Discrete mathematics course assessment - a case study
2010	[23]	Numeracy competence requirements for admission to undergraduate degree programmes: A case study of a programme to prepare pre-registration nursing student candidates for a numeracy entrance test

- Scopus: (difficulty OR difficulties OR problem OR fail OR challenge) AND (mathematics OR mathematical) AND ("secondary school" OR "primary school" OR "high school" OR "middle school" OR "elementary school") AND ("questionnaire" OR exam OR grade OR outcome) AND (algebra OR arithmetic OR fraction OR decimal OR equation OR counting OR set).

The application of the research protocol recovered 5,764 papers. In the end, 18 papers were selected for complete analysis. The Table III shows the results with the final synthesis data extraction.

Table III
SYNTHESIS DATA EXTRACTION - STRING TWO

Database	Total	Inclusions	Exclusions
ACM	41	0	41
IEEE Xplore	5,379	18	5,361
Scopus	344	0	344
Total	5,764	18	5,746

After applying the quality criteria, only 1 paper was selected, as can be seen in the Table IV.

Strategy 3: This string search was executed, yet added no new publications to this review.

- ACM: (difficulties difficulty) +(algebra arithmetic) +"higher education"

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Table IV
ARTICLES LIST - STRING TWO

Year	Reference	Title
2015	[6]	Is mathematical background crucial to freshmen engineering students?

- IEEE Xplore: (((difficulties OR difficulty) AND algebra OR arithmetic) AND "higher education")
- Scopus: (difficulties OR difficulty) AND (algebra OR arithmetic) AND "higher education"

The application of the research protocol recovered 1,449 papers, as can be seen on Table V. During the analysis, 1 paper fitted the including criteria, however it was disregarded after the application of E1 of the exclusion criteria, for the the same work [16] had already appeared in the the final search, using string number two.

Table V
SYNTHESIS DATA EXTRACTION - STRING THREE

Database	Total	Inclusions	Exclusions
ACM	23	0	23
IEEE Xplore	825	0	825
Scopus	601	0	601
Total	1,449	0	1,449

C. Reporting

The three database searches returned a total of 9,836 papers, of with 12, only 0.12%, passed the quality criteria to be analyzed in this SRL. Half of them were conference papers and the other half were journal articles. Figure 1 presents the total of search results in each database in relation to their selected papers.

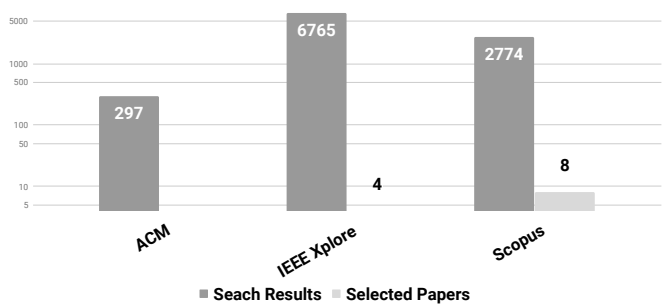

Figure 1. Relation between search results and selected papers in each database.

The selected papers were read in its entirely in order to extract answers to the research questions and the collected data was analyzed and condensed into answers to the research questions, as can be seen as follows:

Q1. Which mathematical topics are most frequently pointed as hard to learn?

Most of the selected papers address more than one mathematical topic, and these can vary in degree of specificity in each paper. To optimize the analysis and reporting of the results, articles were grouped according to its most recurrent subjects.

Among the several mathematics concepts pointed as challenging to students, the most frequently mentioned were fractions [6], [23], [18], [16], [14], functions [14], [17], [19], [18] and linear equations [22], [6], [16]. The relation between them can be seen in Figure 2.

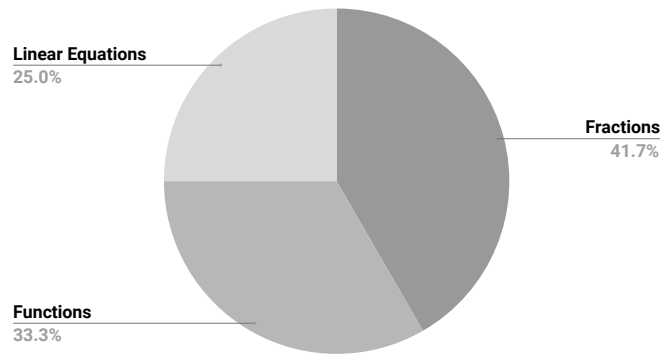

Figure 2. Most difficult basic mathematics topics to students attending higher education.

It is important to emphasize that there are other basic mathematics subjects addressed simultaneously to those grouped in this research question. The topics varied from rational expression and logarithms [22], [6], factorial [13], trigonometry [18], geometry [19] to simple subjects such as proportion [20], decimal numbers [23], [21] and mathematical notations [15].

Q2. Which knowledge subarea is impacted by this deficiency in fundamental mathematical topics?

Most of the researches were, by far, conducted focusing on STEM courses [13], [14], [6], [15], [16], [17], [19], [22]. The other areas were nursing [21], [23], psychology [20] and what was said to be conducted on "undergraduate university studies where mathematics is not a core subject, but still is important for coping with core subjects" [18], as shown in Figure 3.

Papers that did not deal with STEM courses tended to point to simpler subjects. In psychology, the paper pointed to proportion as a problem [20], while in nursing, it addressed deficiencies on applying simple numeracy in practical situations such as performing nutritional assessments, fluid balance and body mass index calculations [19].

As STEM courses, engineering, computer science and mathematics appeared in the researches, often not being specific to which engineering degree exactly. Also, specific disciplines such as calculus [6] and discrete mathematics [22] were mentioned.

Q3. What are the approaches that can help to minimize these high failure rates?

The most widely used approach to minimize the high failure rates, as can be seen in Figure 4, is a change in the

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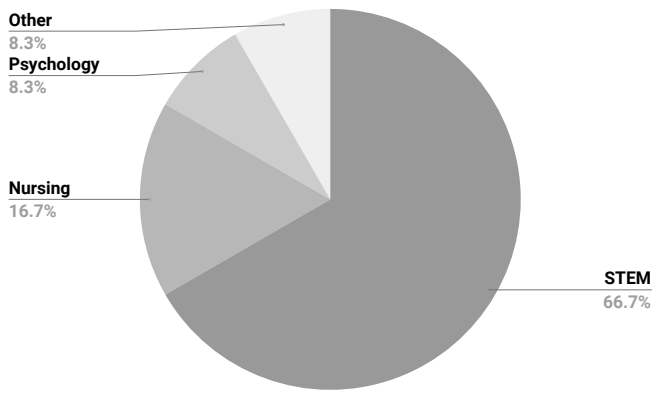

Figure 3. Impacted subareas.

methodology, followed by the development of institutes or programs and, lastly, the use of digital games or tools - with 7, 3 and 2 papers, respectively.

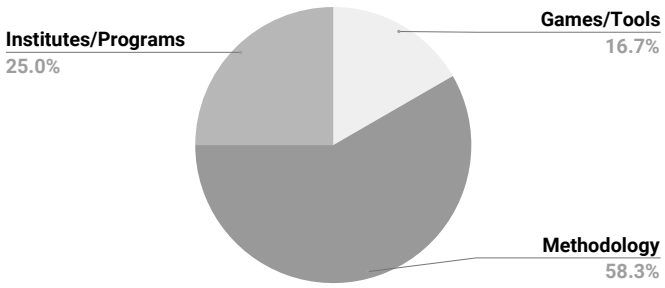

Figure 4. Proposed approaches to help students facing mathematical difficulties.

The papers belonging to each category are presented in Table VI.

Table VI
RELATION BETWEEN PAPERS AND PROPOSED APPROACHES

Category	Reference	Total
Games/Tools	[18], [13]	2
Institutes/Programs	[16], [19], [23]	3
Methodology	[14], [15], [17], [20] , [21], [22], [6]	7

Papers in the *Games/Tools* category, propose the use of digital tools, such as games, as means to improve the students' performance in basic mathematics. They aim to, though students' feedback, develop and test digital tools to support teaching and learning of basic mathematics in higher education.

Papers categorized as *Institutes/Programs* propose the implementation of institutions or projects aimed to support mathematics, such as spaces in which students can receive free, additional help with mathematical topics of their choosing.

Most papers belong to the *Methodology* category, as can be seen in Figure 4. These articles focus on developing initial tests as means to understand the student's mathematical background and presenting tutoring courses or workshops to level up the learner upon entering the course, or during it. They speculate that the high failure rates are caused by the lack of mathematical background, the structure of the class, methodologies adopted by the teacher or the lack of academic commitment.

Q4. What are the technologies proposed to enhance the mathematical learning process?

This question was developed in order to identify how technology and mathematics can work together, however, contrary to what was expected, only two of the articles proposed digital tools to aid in the problem.

One paper presents a diagnosing tool (unnamed) developed to identify nursing students that demand of additional support in mathematics and the areas in which most of them had difficulties [21]. Similarly, the other introduces a game (Count With Me!) with the purpose to help, as they said, "fill in the knowledge gaps" before learning advanced mathematics concepts, tacking the topics of addition, multiplication, factorial, and permutation [13].

As can be seen in Figure 5, a simplified version of Figure 4, only 16.7% of the articles resort to games and digital tools as opposed to non-technological measures to improve the situation.

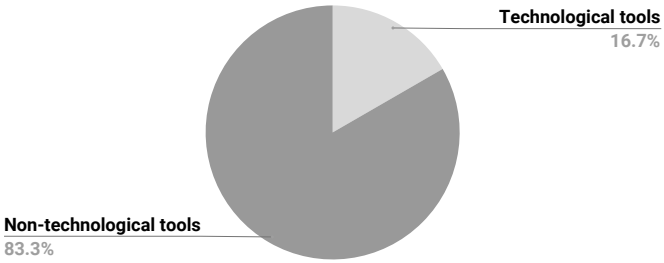

Figure 5. Technology as a way to help mathematics students.

IV. MAIN FINDINGS/DISCUSSIONS

The main findings of our research are listed below:

A. We need more research on difficulties in basic mathematics in higher education

This SRL was conducted using a three phase strategy, each with a different search string, as can be seen in section III. In order to execute the searches, 3 databases consolidated in the scientific scope (ACM, IEEE Xplore, Scopus) were used. These searches returned a total of 9,836 articles, with only 12 being selected. This shows that, although mathematical research is being largely done, few are related to the topic.

Research on this topic is important because, as some of the selected articles show, students with most solid mathematical background tend to perform better. Also, students with persistent difficulties in mathematics have higher failure rates [6] and, consequently, more difficulty in mathematics in higher education.

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B. Proposals to solve this problem when students already reached higher education

The most used approaches to minimize the high failure rate in mathematics in higher education are related to teaching methodology, according to the subsection III-C. The articles [17], [14] demonstrate the need for university professors to change their teaching strategies, taking into account the students' mathematical deficiency. This is because tasks based on mathematical thinking usually are time consuming and require brainstorming.

For other researchers [23] [19], the most appropriate way to supply basic mathematics gaps in higher education is the development of support centers in mathematics, in which low-development students can be reinforced in fundamental mathematics. In general, there is a recommendation for the community to invest in the improvement of teaching methods in basic mathematics.

C. We need to develop different teaching strategies on fundamental mathematics

This question goes beyond changing or introducing new teaching methodologies in the scope of basic mathematics. As can be seen in III-C, only 2 articles presented digital tools as a way to solve this deficiency. This demonstrates the low use of technology as an educational tool for this purpose.

The use of technology in the educational context is fundamental. For years science has been proving the efficiency of the relationship between technology and education. In this context, software, platforms etc. become important actors in terms of improving the teaching of mathematics in higher education.

D. The need to take into account the social formation of the student, such as what they have maintained in elementary education

In [6], one can speculate that the lack of mathematical background, the methodologies adopted by the professor, the structure of the class, the lack of academic commitment are factors that may cause the higher failure rates.

These [20] help build the case that experiences with the operational patterns are a potential source of children's persistent difficulties with mathematical equivalence.

E. STEM is not the only area impacted by the difficulties in basic mathematical concepts in higher education

Deficiency in basic mathematical concepts is a general problem for all areas of knowledge. However, this systematic review shows that the health area is the second most impacted by this problem. To the point of institutions creating support centers focused on mathematics for their students [23].

F. New research questions in the context of difficulties in basic mathematics during higher education

To better understand the problem, it is necessary to think about new research questions that will help to understand the context of these difficulties. Seeking to find, mainly, how, where and why did they emerge? So you can systematically develop methods to minimize them. This systematic review put what researchers have classified as crucial issues to begin to understand the problem, but we know that it is a long way to be solved.

V. Conclusions

The goal of this research is to present the main concepts of basic mathematics in which students face more difficulties during higher education.

In order to follow the standards of a SRL, only publications fitting all criteria were selected for closer inspection. In the end, only twelve papers were selected, pointing out to a scarce amount of literature available on the topic, which demonstrates the need to perform more research on it.

The review presented the topics most frequently identified as challenging to students, and the subareas impacted by it. Also, it presented the techniques used to minimize this deficiency and the technologies proposed to improve this scenario.

Students having difficulties performing simple arithmetic procedures (addition, subtraction, multiplication, division and exponentiation) will, eventually, reach algebraic procedures, which requires previous arithmetic knowledge, and in this scenario, the problem starts to take form.

In addition, research shows that the basic concept that presents the greatest difficulty among students is fraction. Which corroborates with Augustus De Morgan, writing in [24] acknowledges that the learning of fractions is expected to "present extraordinary difficulties". This was true in the nineteenth century and it is still true today.

The paper [6] shows that good mathematical background is crucial for a good student performance in a discipline like calculus. It also presents the high rates of disapproval and evasion that results in student retention in the first semester of the course, evasion of the course in general and, in the worst cases, total dropout from the university.

There is, also, a shortage of proposals to minimize the deficiency faced by these students, which is a problem as this scenario can't be blamed solely on the students, being a consequence of the educational system.

In a way, it is responsibility of all the people composing the educational complex, consequently being a problem of society as a hole. However, the actors of education, the people who are directly involved in the process, have an obligation to seek solutions.

VI. Acknowledgments

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